

CO1 - AT-1 - SHORT ANSWER TEST 192524311

Slot D - DSA0405 - Fundamentals of Data Science July 11th 2026

1. What is Data Science? Explain its primary objective.

- * Data Science is the field of extracting knowledge and insights from data using scientific methods, statistics and computing.
- * It combines data analysis, machine learning and domain expertise to support decision making and solve realworld problems.

2. Why is Data Science important in today's digital world? Mention any 4 reasons.

Data Science helps organisations make data-driven decisions. It improves efficiency through automation and predictive analysis. It enables personalisation innovation and better custom experiences.

3. List any 4 benefits of Data Science in business and society.

Data Science improves decision-making and increases operational efficiency. It helps predict trends reduce risks and improve public services.

4. Mention 4 real world applications of Data Science.

Data Science is used in healthcare, finance, e-commerce and weather forecasting. It helps in disease prediction, fraud detection, recommendations and climate analysis.

5. What are the different facets of data? Briefly explain each.

The main facets of data are qualitative, quantitative, structured and unstructured. They represent descriptive information, numerical values, organised data and non-organised data.

6. Differentiate between structured, semistructured and unstructured data with suitable examples.

Structured data is organised in tables (eg. databases), semi-structured data has partial organization (eg. JSON, XML).

Unstructured data has no fixed format such as images, videos and emails.

7. What is Big-Data? State its key characteristics.

Bigdata refers to large and complex datasets that cannot be handled by traditional tools. Its 5Vs are Volume, Velocity, Variety, Veracity and Value.

8. Explain The Big data ecosystem. Name any 4 technologies used in it.

The Bigdata ecosystem includes Tools for storing, processing and analyzing large datasets.

Examples : Hadoop, Spark, Hive and Base.

9. What are the major phases of the Data Science process?

The major process phases are data collection, cleaning, exploration, modeling and evaluation. These steps convert raw data into meaningful insights and decisions.

10. What is data retrieval? Mention 2 common sources from which data can be retrieved.

Data retrieval is the process of correcting errors and improving data quality. Techniques include removing duplicates, handling missing values, correcting inconsistencies and removing outliers.

11. Why is data collection considered an important step in the data science process?

Data collection provides the information foundation for analysis and model building. Accurate and relevant data leads to reliable results and insights.

12. What is data cleansing? List any 4 common data cleansing techniques.

Data cleansing is the process of accessing and obtaining data for analysis. It includes removing duplicates, handling missing values and removing outliers.

13. What is data integration? Why is it necessary in Data Science?

Data integration combines data from different sources into a single dataset. It ensures consistency and provides a complete view for analysis.

14. Explain data transformation with suitable examples.

Data transformation converts data into a suitable format by analysis. Examples include converting data formats, normalization, and encoding categorical values.

15. What is data analysis? Mention any 4 techniques used for data analysis.

Data analysis is the process of examining data to extract useful information and insights. Common techniques are descriptive, diagnostic, predictive and prescriptive analysis.

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